

LINUX DIRECTORY STRUCTURE



+
+
+
+



Created By DevOps Shack

[Check Out Batch-9 Of DevSecOps & Cloud DevOps Bootcamp](#)

Linux Directory Structure – The Ultimate Guide

Ever wondered **what's inside the Linux filesystem** and **why each directory exists?** Understanding the **Linux directory structure** is **critical** for any DevOps engineer, SysAdmin, or Cloud professional! 🏆

Let's **break it down** and explore the **core directories** you'll work with daily!

/ (Root) – The Foundation of Linux

Everything in Linux starts from /. It is the **top-most directory** and **all other directories** are inside it.

/bin – Essential User Binaries

✦ Contains essential **binary executables** like ls, cp, mv, cat, grep.

✦ Needed for the system to **boot and run basic commands**.

✓ Try this:

```
ls /bin
```

/sbin – System Binaries

✦ Contains system binaries for **administrative tasks** like shutdown, fdisk, iptables, mount.

✦ Only root users can execute most commands here.

✓ Try this:

```
ls /sbin
```

/usr – User Programs & Utilities

✦ Contains **user-installed software**, libraries, and documentation.

✦ /usr/bin → Non-essential system binaries

✦ /usr/sbin → System admin binaries

✦ /usr/local → Locally installed software

✓ Check installed binaries:

```
ls /usr/bin
```

/etc – System Configuration Files

 Contains **all configuration files** and startup scripts.

 Examples:

- ◆ /etc/passwd → User account details
- ◆ /etc/fstab → Filesystem mounts
- ◆ /etc/hosts → Hostname resolution

 View system users:

```
cat /etc/passwd
```

/home – User Home Directories

 Contains **personal files, configs, and settings** for users.

 Example: /home/username contains files for username.

 List all user home directories:

```
ls /home
```

/var – Variable Data Files

 Stores **log files, temporary files, cache, databases**.

 Important directories:

- ◆ /var/log → Logs (e.g., /var/log/syslog, /var/log/auth.log)
- ◆ /var/lib → Databases, package manager data
- ◆ /var/tmp → Temporary files

 View logs:

```
cat /var/log/syslog
```

/tmp – Temporary Files

 Used for **temporary storage**, cleared on reboot.

 Example: Applications store temporary files here.

 Create a temp file:

```
touch /tmp/testfile
```

/dev – Device Files

✦ Contains files representing **devices** like hard disks, USB drives, and terminals.

✦ Examples:

- ◆ /dev/sda1 → First partition of the first hard disk
- ◆ /dev/null → Discard anything written to it
- ◆ /dev/tty → Terminal devices

✓ Check available disks:

```
ls /dev/sd*
```

/mnt & /media – Mount Points

✦ /mnt → Temporary mount point for external storage.

✦ /media → Auto-mounted external drives.

✓ Mount a drive:

```
mount /dev/sdb1 /mnt
```

/opt – Optional Software Packages

✦ Stores **third-party software** installed manually.

✓ Example:

```
ls /opt
```

/proc & /sys – Kernel & System Information

✦ /proc contains **runtime system information**, e.g., CPU, memory.

✦ /sys contains **kernel-related system files**.

✓ Check CPU info:

```
cat /proc/cpuinfo
```